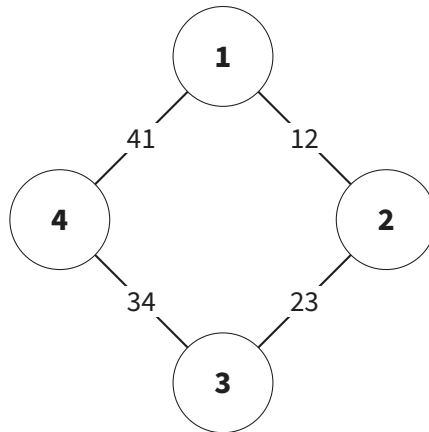


Which pictures are possible?

Name: _____

Date: _____

Start all lamps OFF. Press a line to flip **both** end lamps. Dark means ON. A lamp already ON turns OFF. Write 12 for the line joining lamps 1 and 2.



1. Three targets.

Reset to all OFF before each target. Find moves, or give a reason the target cannot happen.

ON lamps	My moves or my reason
1 and 2	
1 and 3	
1 only	

2. Look at one move closely.

The two end lamps might both be OFF, both ON, or one of each. How does the number of ON lamps change in these three cases?

A rule that survives every move:

Try it with objects. Explain by pointing, talking, or drawing.

Find the whole world

3. Make a complete collection.

Draw every reachable picture in the boxes: a dot for each lamp, filled if ON. Keep the order 1, 2, 3, 4. The all-OFF picture counts. There may be more boxes than you need.

Our count: _____ How do you know none are missing?

4. Make two far-apart lamps.

Choose any two lamps. Trace a path between them. Press each line on your path once. What happens to the lamps in the middle?

Explain why a path changes only its two ends.

Go further: What if the ring has 5 lamps? Can every picture with an even number of ON lamps still be made? Pair its ON lamps and try paths.

Try it with objects. Explain by pointing, talking, or drawing.